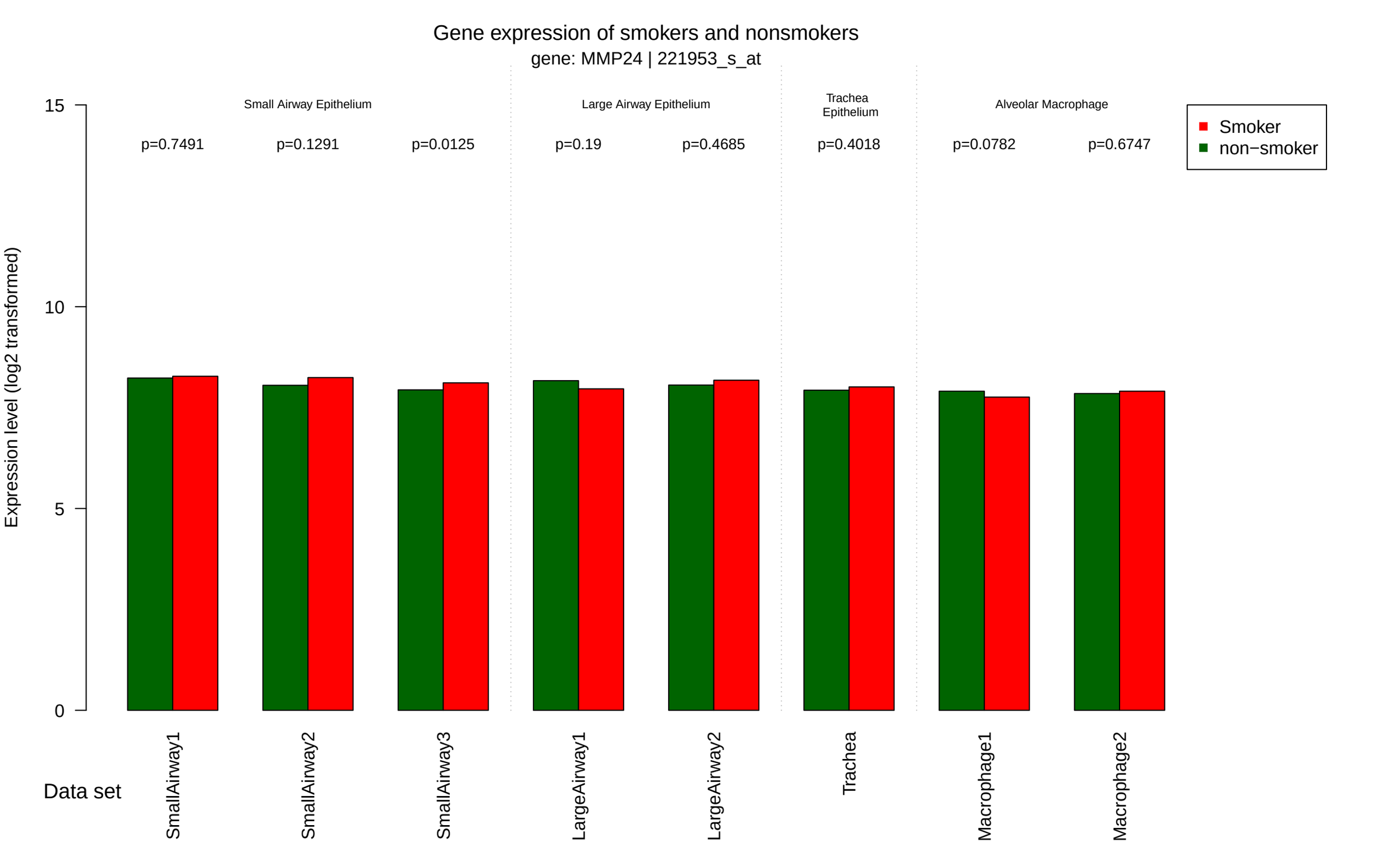
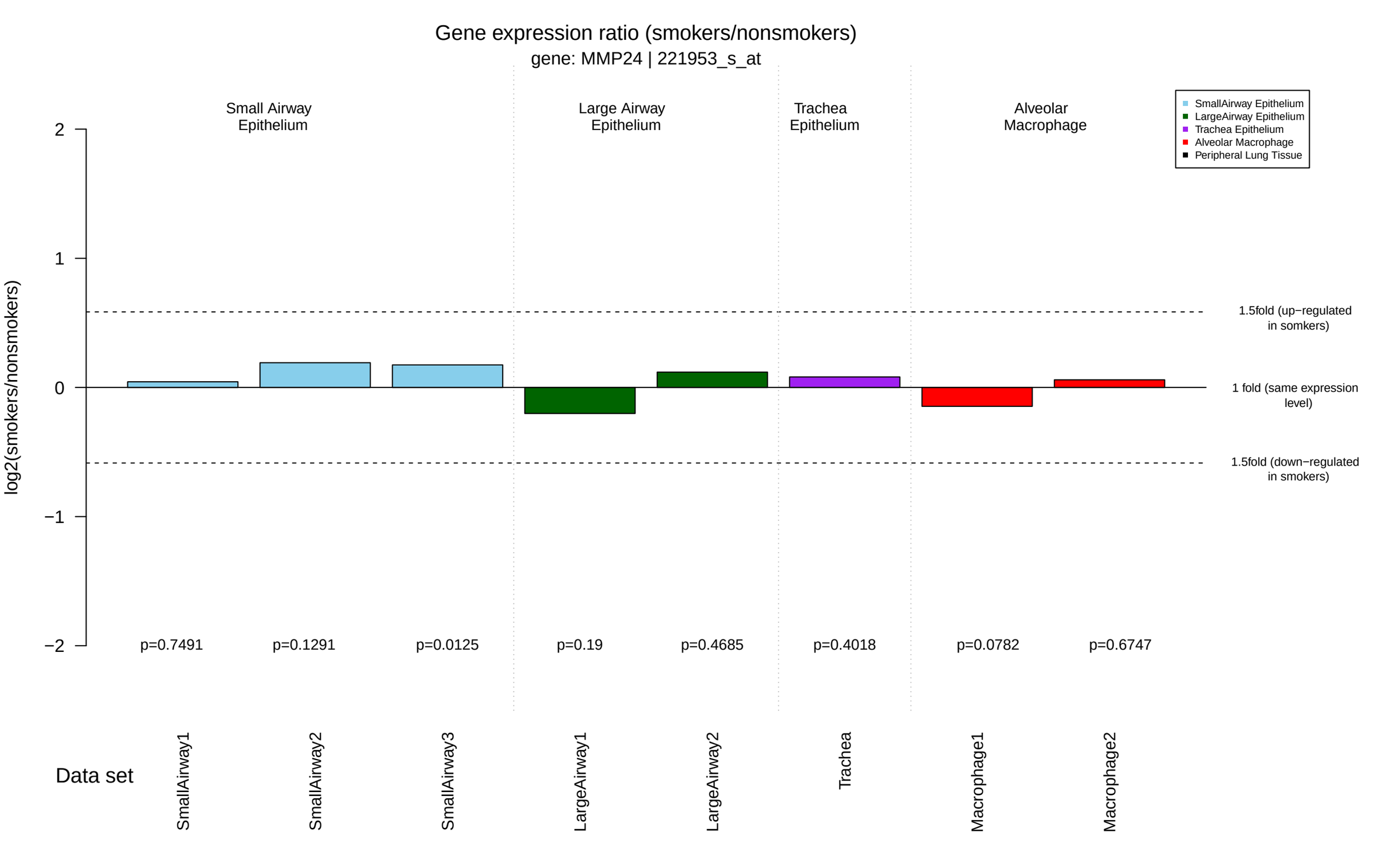
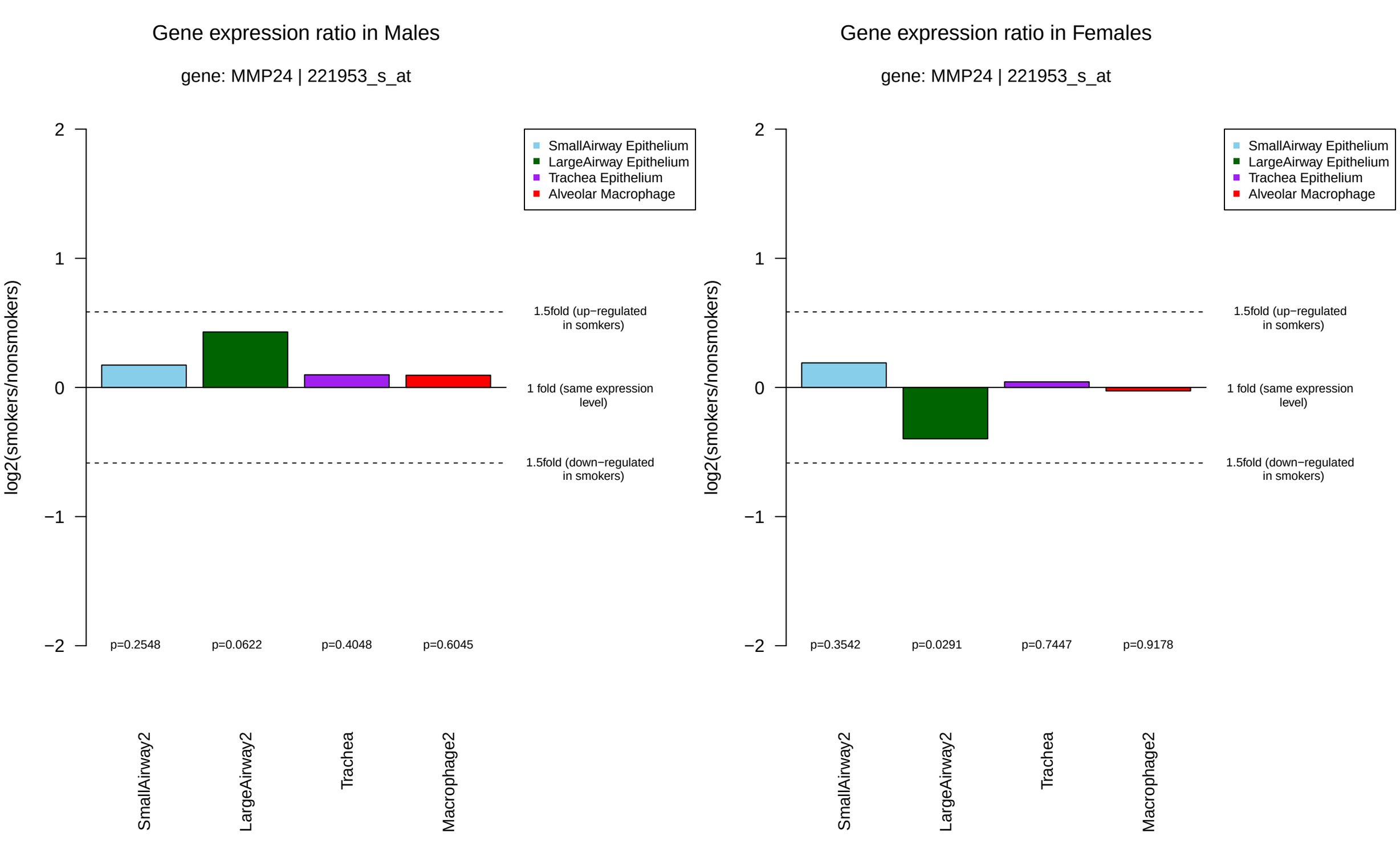
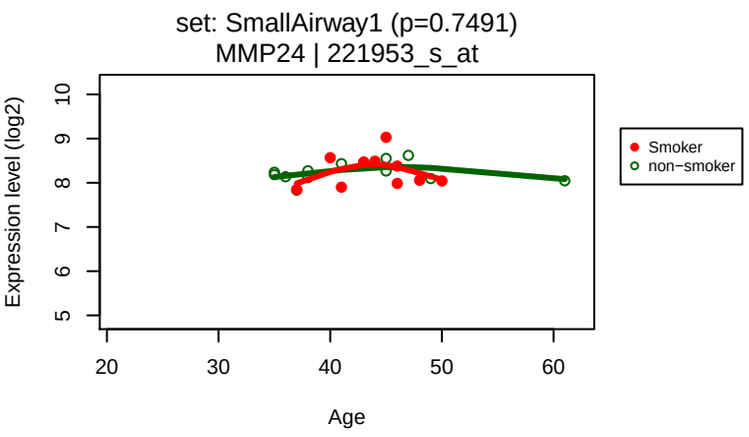
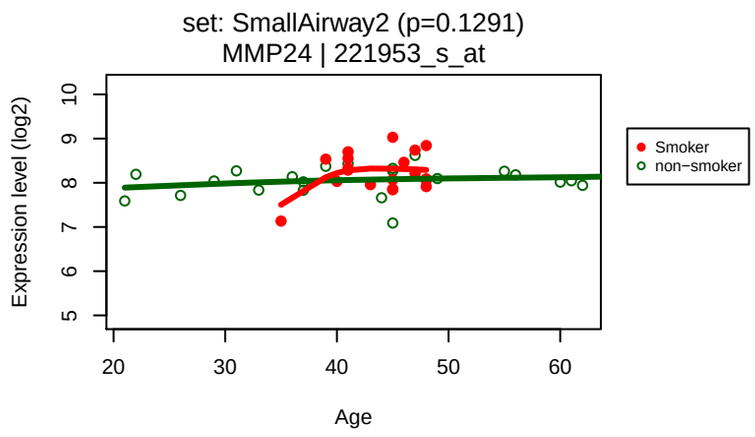
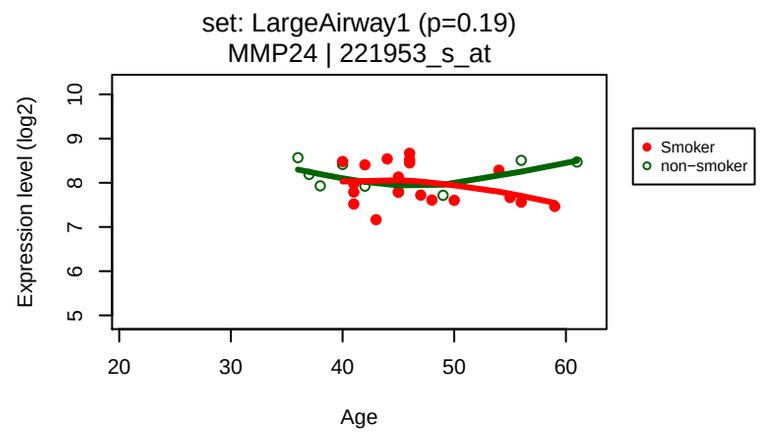

Gene expression of smokers and nonsmokers
gene: MMP24 | 221953_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7491
p=0.1291
p=0.0125
p=0.19
p=0.4685
p=0.4018
p=0.0782
p=0.6747
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

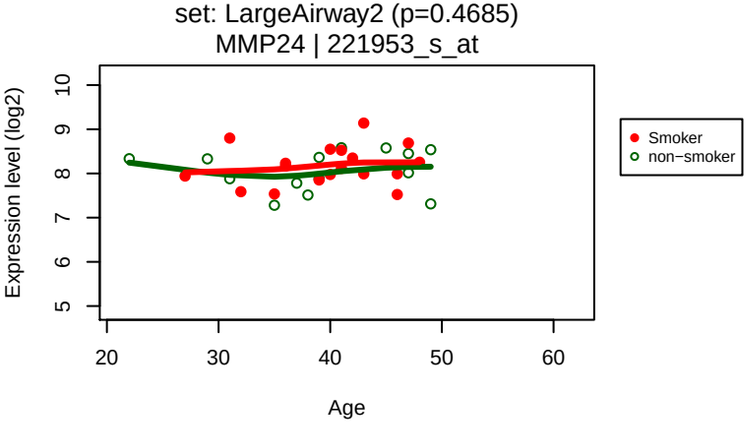

Gene expression ratio (smokers/nonsmokers)
gene: MMP24 | 221953_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7491
p=0.1291
p=0.0125
p=0.19
p=0.4685
p=0.4018
p=0.0782
p=0.6747
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

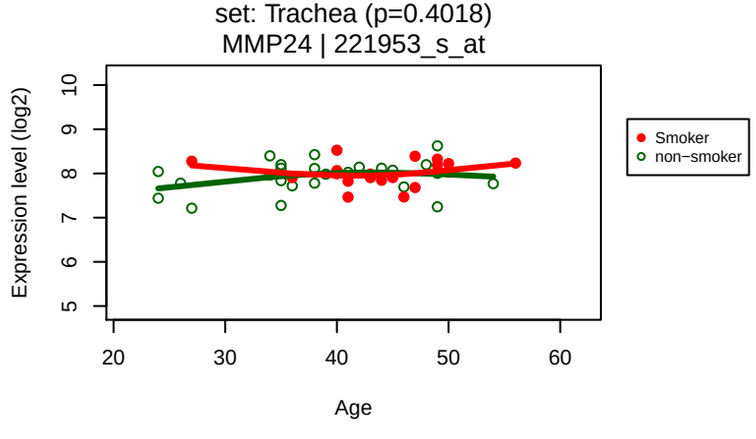

Gene expression ratio in Males
gene: MMP24 | 221953_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2548
p=0.0622
p=0.4048
p=0.6045
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP24 | 221953_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3542
p=0.0291
p=0.7447
p=0.9178
SmallAirway2
LargeAirway2
Trachea
Macrophage2

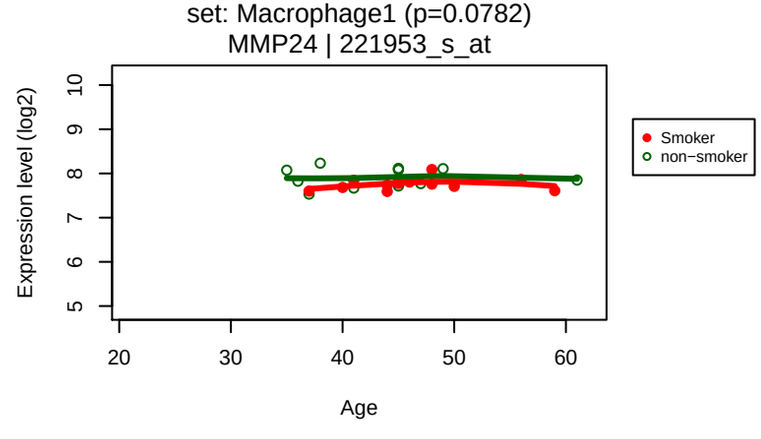

set: SmallAirway1 (p=0.7491)
MMP24 | 221953_s_at
Expression level (log2)
Age
Smoker
non−smoker

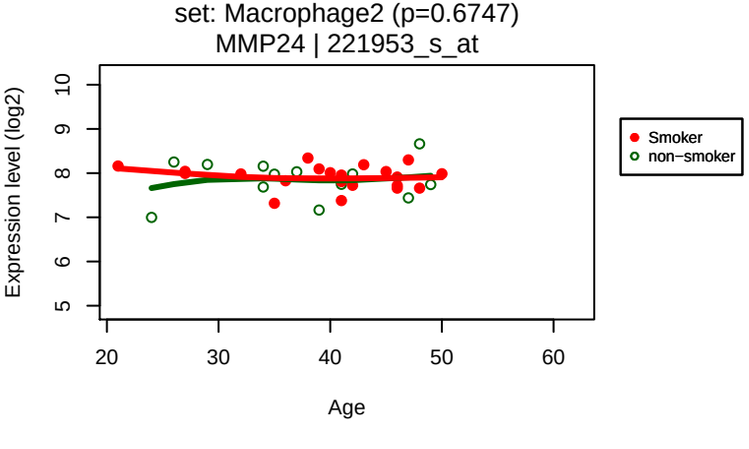

set: SmallAirway2 (p=0.1291)
MMP24 | 221953_s_at
Expression level (log2)
Age
Smoker
non−smoker

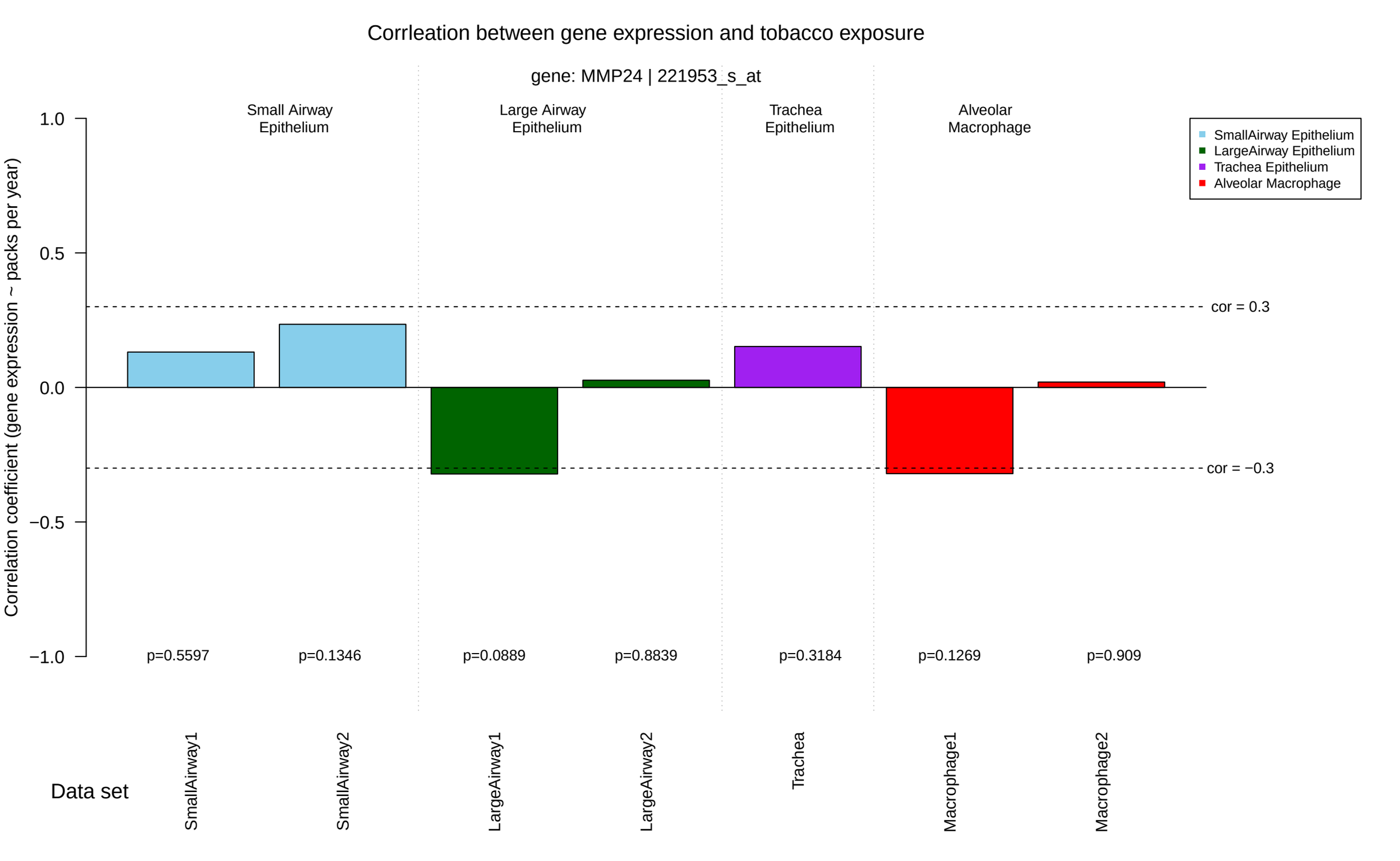

set: LargeAirway1 (p=0.19)
MMP24 | 221953_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4685)
MMP24 | 221953_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4018)
MMP24 | 221953_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0782)
MMP24 | 221953_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6747)
MMP24 | 221953_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP24 | 221953_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5597
p=0.1346
p=0.0889
p=0.8839
p=0.3184
p=0.1269
p=0.909
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2